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Lecture 1: Optimal Transport

Lecture 2: Generative Flows

Lecture 3: From 1D Processes to Flows

Lecture 4: Bayesian Inverse Problems

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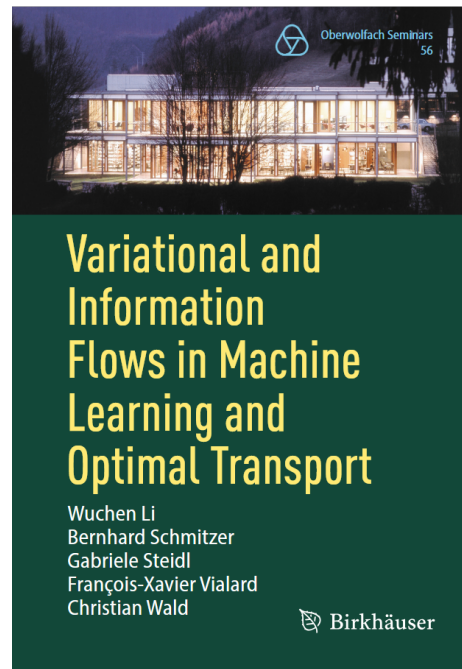
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2. Generative Flows

1. Basic Idea of Generative Flows
2. Absolutely Continuous Curves in $(\mathcal{P}_2(\mathbb{R}^d), W_2)$
3. Flow Matching
4. A Glimpse to Score-based Diffusion



Refs: Wald/St.: Flow Matching: Markov kernels, transport plans and stochastic processes, MFO Seminar Series, Springer 2025

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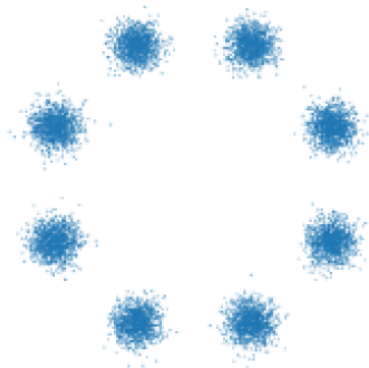
1. Basic Idea of Generative Flows

Aim: Sampling from a probability distribution μ_X (target distribution) given:

- ◆ some samples (oft many = data hungry)
- ◆ probability density function $p_X(x) = \exp(-\psi(x))/C$ (Boltzmann density) with or without normalizing factor C

It is only easy to sample from

- ◆ 1d distributions (cdf)
- ◆ $d \gg 1$ uniform or Gaussian "friendly" distributions μ_Z (latent distribution)



$$d = 2$$

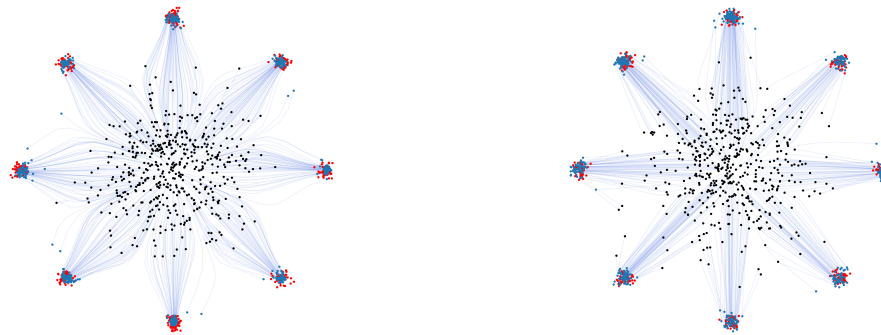


$$d = 28^2 = 784$$

Idea: Learn a curve from μ_Z to μ_X

Basic methods:

- ◆ score-based diffusion models (Sohl-Dickstein 2015; Song/Ermon et al. 2019)
- ◆ flow matching (Lipman et al. 2022/23, Liu et al. 2022/23, Albergho et al. 2023)
- ◆ consistency models, e.g. inductive moment matching (Zhou et al. ICLM 2025, Boffi et al. 2025)



Trajectories of points via flow matching from different couplings, $d = 2$



Single trajectory for a flow matching model trained on cat images, $d = 256^2$

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In a complete metric space (X, d) , a curve $\gamma : I \rightarrow X$ is called **absolutely continuous** ($AC^2(\mathbb{R}^d)$), if there exists as $g \in L_2(I)$ such that

$$d(\gamma(s), \gamma(t)) \leq \int_s^t g(r) \, dr \quad \text{for } s \leq t, \, s, t \in I.$$

Then there exists a.e. the metric derivative (speed) which is indeed the smallest function g in the above inequality

$$|\gamma'| (t) := \lim_{s \rightarrow t} \frac{d(\gamma(t), \gamma(s))}{t - s}.$$

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- ◆ A narrowly continuous curve $\mu_t : I = [0, 1] \rightarrow \mathcal{P}_2(\mathbb{R}^d)$ **absolutely continuous**, if there exists a Borel measurable vector field $v : I \times \mathbb{R}^d \rightarrow \mathbb{R}^d$ with $\|v_t\|_{L_2(\mu_t)} \in L_2(I)$ such that (μ_t, v_t) satisfies the **continuity equation (CE)**

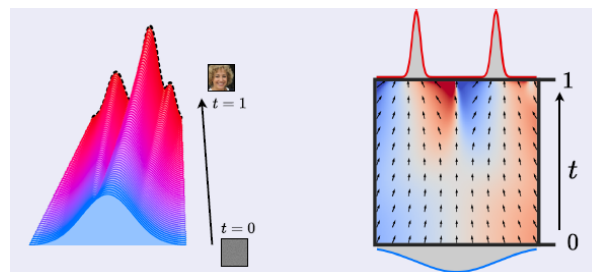
$$\partial_t \mu_t + \nabla_x \cdot (\mu_t v_t) = 0$$

- ◆ If μ_t has a smooth density p_t , then

$$\partial_t p_t + \nabla_x \cdot (p_t v_t) = 0$$

- ◆ otherwise (CE) is meant in a weak sense, i.e. for all $\varphi \in C_c^\infty((a, b) \times \mathbb{R}^d)$:

$$\int_I \int_{\mathbb{R}^d} \partial_t \varphi \, d\mu(t) dt + \int_I \int_{\mathbb{R}^d} \langle \nabla_x \varphi, v \rangle \, d\mu_t(x) dt = 0$$



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◆ Let (μ_t, v_t) fulfill (CE) and

$$\int_I \sup_{x \in B} \|v_t(x)\| + \text{Lip}(v_t, B) \, dt < \infty \quad \forall \text{compact } B \subset \mathcal{B}(\mathbb{R}^d).$$

Then there exists a solution $\phi : I \times \mathbb{R}^d \rightarrow \mathbb{R}^d$ of the **ODE**

$$\partial_t \phi(t, x) = v_t(\phi(t, x)), \quad \phi(0, x) = x$$

and $\phi(t, \cdot)_\# \mu_0 = \mu_0 \circ \phi(t, \cdot)^{-1} = \mu_t$.

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Also the other direction is true under some conditions.

Assume that ϕ satisfies the ODE for some locally bounded vector field v_t on $I = (a, b)$. For $\varphi \in C_c^\infty((a, b) \times \mathbb{R}^d)$, we can compute

$$\begin{aligned} 0 &= \varphi(b, \phi(b, x)) - \varphi(a, \phi(a, x)) \\ &= \int_{\mathbb{R}^d} \varphi(b, \phi(b, x)) - \varphi(a, \phi(a, x)) d\mu_0 \\ &= \int_{\mathbb{R}^d} \int_a^b \frac{d}{dt} (\varphi(t, \phi(t, x))) dt d\mu_0 \\ &= \int_{\mathbb{R}^d} \int_a^b \langle \nabla_x \varphi(t, \phi(t, x)), v_t(t, \phi(t, x)) \rangle + (\partial_t \varphi)(t, \phi(t, x)) dt d\mu_0 \\ &= \int_a^b \int_{\mathbb{R}^d} \langle \nabla_x \varphi, v_t \rangle + \partial_t \varphi d[\phi(t, \cdot)_\# \mu_0] dt \end{aligned}$$

Thus $\mu_t := \phi(t, \cdot)_\# \mu_0$ satisfies (CE) (in a weak sense).

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- ◆ A narrowly continuous curve $\mu_t : I = [0, 1] \rightarrow \mathcal{P}_2(\mathbb{R}^d)$ **absolutely continuous**, if there exists a Borel measurable vector field $v : I \times \mathbb{R}^d \rightarrow \mathbb{R}^d$ with $\|v_t\|_{L_2(\mu_t)} \in L_2(I)$ such that (μ_t, v_t) satisfies the **continuity equation (CE)**

$$\partial_t \mu_t + \nabla_x \cdot (\mu_t v_t) = 0$$

- ◆ Let (μ_t, v_t) as above and

$$\int_I \sup_{x \in B} \|v_t(x)\| + \text{Lip}(v_t, B) \, dt < \infty \quad \forall \text{compact } B \subset \mathcal{B}(\mathbb{R}^d).$$

Then there exists a solution $\phi : I \times \mathbb{R}^d \rightarrow \mathbb{R}^d$ of the **ODE**

$$\partial_t \phi(t, x) = v_t(\phi(t, x)), \quad \phi(0, x) = x$$

and $\phi(t, \cdot)_{\#} \mu_0 = \mu_0 \circ \phi(t, \cdot)^{-1} = \mu_t$.

- ◆ If a nice person 😊 gifts you such a velocity field v_t , then use your favorite ODE solver to transport samples from $\mu_0 = \mu_X$ to samples of $\mu_1 = \mu_Z$ **or backwards** using $-v_{1-t}$.

All you need is v_t !

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How to Learn a Velocity Field?

Task: Learn/approximate v_t by neural network v_t^θ !

Loss function for learning:

$$\mathcal{L}(\theta) := \mathbb{E}_{t \sim \mathcal{U}(0, T), x \sim \mu_t} \left[\|v_t^\theta(x) - v_t(x)\|^2 \right],$$

not accessible 😞.

Idea: Make it **conditional** since up to a constant

$$\mathcal{L}(\theta) = \mathbb{E}_{t \sim \mathcal{U}(0, T), y \sim \mu_1, x \sim \mu_t(\cdot|y)} \left[\|v_t^\theta(x) - v_t^y(x)\|^2 \right]$$

😊 We will see that conditional velocity fields v_t^y belonging to a conditional flow $\mu_t(\cdot|y)$ starting in δ_y are available in an analytic form in certain cases e.g. from couplings of μ_X and μ_Z (optimal transport)

All you need is conditional v_t^y !

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Rewriting the Loss Function

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Random variables: $X_0 \sim P_{X_0} = \mu_0$, $X_1 \sim P_{X_1} = \mu_1$, $X_t \sim P_{X_t} = \mu_t$, $\mu_t(\cdot|y) = P_{X_t|X_1=y}$

$$P_{X_t, X_1} = P_{X_t|X_1=y} \times_y P_{X_1} = P_{X_1|X_t=x} \times_x P_{X_t}$$

Theorem. Let $(\mu_t(\cdot|y), v_t^y)$ satisfy the (CE), then the following (μ_t, v_t) also satisfy the (CE):

$$\mu_t = \pi_{\#}^1(\mu_t(\cdot|y) \times_y \mu_1), \quad p_t(x) = \int p_t^y(x) d\mu_1(y) \quad (1)$$

$$v_t(x) = \int_{\mathbb{R}^d} v_t^y(x) dP_{X_1|X_t=x}(y) \quad (2)$$

Rewriting the loss function:

$$\mathcal{L}(\theta) := \mathbb{E}_{t \sim \mathcal{U}(0, T), x \sim \mu_t} \left[\left\| v_t^\theta(x) - v_t(x) \right\|^2 \right] = \mathbb{E} \left[\|v_t^\theta\|^2 \right] - 2\mathbb{E} \left[\langle v_t, v_t^\theta \rangle \right] + \underbrace{\mathbb{E} \left[\|v_t\|^2 \right]}_{\text{const}}$$

Consider first and second term separately.

1.

$$\begin{aligned}
 \mathbb{E} \left[\|v_t^\theta(x)\|^2 \right] &= \int_0^1 \int_{\mathbb{R}^d} \|v_t^\theta(x)\|^2 d\mu_t(x) dt \\
 &= \int_0^1 \int_{\mathbb{R}^d} \int_{\mathbb{R}^d} \|v_t^\theta(x)\|^2 dP_{X_1|X_t=x}(y) dP_{X_t}(x) dt \\
 &= \int_0^1 \int_{\mathbb{R}^d \times \mathbb{R}^d} \|v_t^\theta(x)\|^2 dP_{X_t, X_1}(x, y) dt \\
 &= \int_0^1 \int_{\mathbb{R}^d \times \mathbb{R}^d} \|v_t^\theta(x)\|^2 dP_{X_t|X_1=y}(x) dP_{X_1}(y) dt \\
 &= \mathbb{E}_{t \sim \mathcal{U}(0, T), y \sim \mu_1, x \sim \mu_t(\cdot|y)} \left[\|v_t^\theta(x)\|^2 \right]
 \end{aligned}$$

$$\begin{aligned}
 2. \quad \mathbb{E}_{(t, x) \sim \mu_t \times_t dt} \left[\langle v_t^\theta(x), v_t(x) \rangle \right] &= \int_0^1 \int_{\mathbb{R}^d} \left\langle v_t^\theta(x), \int_{\mathbb{R}^d} v_t^y(x) dP_{X_1|X_t=x}(y) \right\rangle dP_{X_t}(x) dt \\
 &= \int_0^1 \int_{\mathbb{R}^d \times \mathbb{R}^d} \langle v_t^\theta(x), v_t^y(x) \rangle dP_{X_t, X_1}(x, y) dt \\
 &= \mathbb{E}_{t \sim \mathcal{U}(0, T), y \sim \mu_1, x \sim \mu_t(\cdot|y)} \left[\langle v_t^\theta(x), v_t^y(x) \rangle \right]
 \end{aligned}$$

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How we can condition velocity fields?

- ◆ from couplings/linear processes
- ◆ from special processes:

Duong, Chemseddine, Friz, Steidl: Telegrapher's generative model via Kac flows

Chemseddine, Kornhardt, Duong, Steidl: Adapting noise to data: generative flows from 1D processes

Conditioned velocity fields from couplings

Curve ends: $\mu_0 = \mu_Z, \mu_1 = \mu_X$

Arbitrary coupling: $\alpha \in \Pi(\mu_0, \mu_1), (X_0, X_1) \sim \alpha$

Interpolation map: $e_t(x, y) = (1 - t)x + ty, t \in [0, 1]$

Theorem The pair (μ_t, v_t) induced by α , i.e.

$$\mu_t := e_{t,\#}\alpha \quad X_t = (1 - t)X_0 + tX_1 \sim \mu_t$$

$$v_t \mu_t := e_{t,\#}[(y - x)\alpha] \quad \text{for a.e. } t \in [0, 1].$$

satisfies the (CE). Further,

$$\mu_t^y = e_{t,\#}(P_{X_t|X_1=y} \times \delta_y),$$

$$v_t^y = \frac{y - x}{1 - t}$$

fulfill the (CE) and are related to μ_t and v_t by (1) and (2), resp.

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Example

Example: Let $\alpha = \mu_0 \times \mu_1$ and $\mu_0 = \mathcal{N}(0, I_d)$. Then (μ_t, v_t) above is given by $\mu_t = p_t dx$:

$$p_t(x) = C_t \int_{\mathbb{R}^d} e^{-\frac{\|x-ty\|^2}{2(1-t)^2}} d\mu_1(y) \quad \text{with} \quad C_t := \left(2\pi(1-t)^2\right)^{-\frac{d}{2}}$$

$$v_t = \frac{1-t}{t} \underbrace{\nabla \log p_t}_{\text{score}} + \frac{x}{t}$$

and thus for $\mu_1 = \delta_y$,

$$p_t(x|y) = C_t e^{-\frac{\|x-ty\|^2}{2(1-t)^2}}$$

$$v_t^y(x) = \frac{y-x}{1-t}$$



and $\mu_t(\cdot|y) = (e_t)_\# \alpha = T_{t,\#}^y \mu_0$ with $T_t^y(x) := (1-t)x + ty$

ODE with $T_t^y = \phi_t$:

$$\partial_t T_t^y(x) = v_t^y(T_t^y(x)) = \frac{y - ((1-t)x + ty)}{1-t} = y - x$$

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Example

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1. For $\mu_t := e_{t,\#}\alpha$ we get with $x := (z - yt)/(1 - t)$:

$$\begin{aligned}\int \varphi \, d\mu_t &= \int \varphi \, de_{t,\#}\alpha = \int \varphi(e_t(x, y)) \, d\alpha(x, y) \\ &= (2\pi)^{-\frac{d}{2}} \int \varphi((1-t)x + ty) e^{-\frac{\|x\|^2}{2}} \, dx d\mu_1(y), \\ &= (2\pi(1-t)^2)^{-\frac{d}{2}} \int \varphi(z) e^{-\frac{\|z-ty\|^2}{2(1-t)^2}} \, d\mu_1(y) dz.\end{aligned}$$

Example continued

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2. Further we verify for $f_t := \frac{1-t}{t} \log p_t + \frac{1}{2t} \|\cdot\|^2$ and $\mu_t = p_t dx$ that

$$\begin{aligned}
 \int_{\mathbb{R}^d} \varphi \nabla f_t d\mu_t &= \frac{1-t}{t} \int_{\mathbb{R}^d} \varphi(x) \frac{\nabla p_t(x)}{p_t(x)} p_t(x) dx + \frac{1}{t} \int_{\mathbb{R}^d} \varphi(x) x p_t(x) dx \\
 &= \frac{1}{t(1-t)^{d+1} (2\pi)^{\frac{d}{2}}} \int_{\mathbb{R}^d \times \mathbb{R}^d} \varphi(x) (ty - x) e^{-\frac{\|x-ty\|^2}{2(1-t)^2}} d\mu_1(y) dx \\
 &\quad + \frac{1}{t(1-t)^d (2\pi)^{\frac{d}{2}}} \int_{\mathbb{R}^d \times \mathbb{R}^d} \varphi(x) x e^{-\frac{\|x-ty\|^2}{2(1-t)^2}} d\mu_1(y) dx \\
 &= \frac{1}{(1-t)^{d+1} (2\pi)^{\frac{d}{2}}} \int_{\mathbb{R}^d \times \mathbb{R}^d} \varphi(x) (y - x) e^{-\frac{\|x-ty\|^2}{2(1-t)^2}} d\mu_1(y) dx.
 \end{aligned}$$

On the other hand

$$\begin{aligned}
 \int_{\mathbb{R}^d} \varphi v_t d\mu_t &= \int_{\mathbb{R}^d} \varphi de_{t,\#}(y - x) \alpha \\
 &= \frac{1}{(2\pi)^{\frac{d}{2}}} \int_{\mathbb{R}^d \times \mathbb{R}^d} \varphi(e_t(x, y)) (y - x) e^{-\frac{\|x\|^2}{2}} d\mu_1(y) dx \\
 &= \frac{1}{(1-t)^{d+1} (2\pi)^{\frac{d}{2}}} \int_{\mathbb{R}^d \times \mathbb{R}^d} \varphi(z) (y - z) e^{-\frac{\|z-ty\|^2}{2(1-t)^2}} d\mu_1(y) dz
 \end{aligned}$$

Recall $\mu_t = P_{X_t} = (e_t)_\# \mu_0 = (e_t)_\# P_{X_0}$, $\alpha = P_{X_0, X_1}$, $v_t^y = \frac{y-x}{1-t}$

$$\begin{aligned} \mathcal{L}(\theta) &:= \mathbb{E}_{t \sim \mathcal{U}(0, T), y \sim \mu_1, x \sim \mu_t(\cdot | y)} \left[\left\| v_t^\theta(x) - v_t^y(x) \right\|^2 \right] \\ &= \int_0^1 \int_{\mathbb{R}^d \times \mathbb{R}^d} \left\| v_t^\theta(x) - v_t^y(x) \right\|^2 dP_{X_t | X_1=y}(x) dP_{X_1}(y) dt \\ &= \int_0^1 \int_{\mathbb{R}^d \times \mathbb{R}^d} \left\| v_t^\theta(x) - v_t^y(x) \right\|^2 dP_{X_t, X_1}(x, y) dt \\ &= \int_0^1 \int_{\mathbb{R}^d \times \mathbb{R}^d} \left\| v_t^\theta(e_t(x, y)) - v_t^y(e_t(x, y)) \right\|^2 dP_{X_0, X_1}(x, y) dt \\ &= \mathbb{E}_{t \sim \mathcal{U}(0, T), (x, y) \sim \alpha} \left[\left\| v_t^\theta(e_t(x, y)) - (y - x) \right\|^2 \right] \end{aligned}$$

All you need is a coupling α !

- ◆ independent coupling: $\alpha = \mu_0 \times \mu_1$
- ◆ optimal coupling: α minimizer of $W_2^2(\mu_0, \mu_1)$ with **advantage** that $\alpha = (\text{Id}, T)_\# \mu_0$ is induced (often) by a Monge map $T : \mathbb{R}^d \rightarrow \mathbb{R}^d$ and we have **straight trajectories** $\phi_t(x) = T_t(x) := (1-t)x + tT(x)$



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Let $X_t \in L^2(\Omega, \mathbb{R}^d, \mathbb{P})$ be continuously differentiable in every $t \in [0, 1]$. Let

$$\mu_t := X_{t,\#}\mathbb{P} = P_{X_t}, \quad \text{and} \quad v_t := \mathbb{E}[\partial_t X_t | X_t = \cdot].$$

◆ (μ_t, v_t) fulfills the continuity equation

Example: Let $X_t = (1 - t)X_0 + tX_1$ for **independent** random variables X_0, X_1 , where $X_0 \sim \mathbb{N}(0, I_d)$. Then $P_{X_t|X_1=y} = C_t e^{-\frac{\|x-ty\|^2}{2(1-t)^2}}$ and we get

$$v_t(x) = \mathbb{E}[\partial_t X_t | X_t = x] = \mathbb{E}[X_1 - X_0 | X_t = x] = \mathbb{E}\left[\frac{X_1}{1-t} - \frac{X_t}{1-t} \middle| X_t = x\right]$$

Using Tweedie's (Miyasawa's) formula

$$\nabla_x \log p_t(x) = \frac{t\mathbb{E}[X_1 | X_t = x] - x}{(1-t)^2}$$

$$\mathbb{E}[X_1 | X_t = x] = \frac{(1-t)^2}{t} \nabla_x \log p_t(x) + \frac{x}{t}$$

we finally get with $\mathbb{E}[X_t | X_t] = X_t$ that

$$\begin{aligned} v_t(x) &= \frac{1-t}{t} \nabla_x \log p_t(x) + \frac{x}{t(1-t)} - \frac{x}{1-t} \\ &= \frac{1-t}{t} \nabla_x \log p_t(x) + \frac{x}{t}, \end{aligned}$$

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Remark: Tweedie's Formula

Consider

$$y = x + \eta, \quad \eta \sim \mathcal{N}(0, \sigma^2 I)$$

The observation density is related to the prior via Bayes' Theorem:

$$p(y) = \int p(y|x)p(x) \, dx = \int g(y-x)p(x) \, dx, \quad g(z) = \frac{1}{(2\pi\sigma^2)^{N/2}} e^{-\|z\|^2/(2\sigma^2)}.$$

For a given observation y , the minimum mean squared error (MMSE) estimator is given as

$$\hat{x}(y) = \int x p(x|y) \, dx = \int x \frac{p(y|x)p(x)}{p(y)} \, dx.$$

The MMSE can be computed using Tweedie's (Miyasawa's) formula

$$\hat{x}(y) = y + \sigma^2 \nabla_y \log p(y),$$

which can be seen directly by computing

$$\nabla_y p(y) = \frac{1}{\sigma^2} \int (x-y) g(y-x) p(x) \, dx = \frac{1}{\sigma^2} \int (x-y) p(y, x) \, dx,$$

and then multiplying both sides by $\sigma^2/p(y)$ gives

$$\frac{\sigma^2 \nabla_y p(y)}{p(y)} = \int x p(x|y) \, dx - \int y p(x|y) \, dx = \hat{x}(y) - y.$$

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Forward SDE computes $(X_t)_{t \in [0, T]}$ by

$$dX_t = f(t, X_t) dt + g(t) dW_t, \quad X_0 \sim P_{\text{data}},$$

where W_t is a standard Brownian motion.

Example: Usual choice is

$$f(t, X_t) := -\frac{1}{2}\beta_t X_t \quad \text{and} \quad g(t) := \sqrt{\beta_t}$$

with a positive, increasing so-called "time schedule" β_t . Then SDE becomes linear

$$dX_t = -\frac{1}{2}\beta_t X_t dt + \sqrt{\beta_t} dW_t, \quad X_0 \sim P_{\text{data}},$$

which has the closed form solution

$$X_t = \sqrt{1 - e^{-h(t)}} Z + e^{-\frac{h(t)}{2}} X_0, \quad h(t) := \int_0^t \beta_s ds, \quad Z \sim \mathcal{N}(0, I_d)$$

Hence, X_t reaches Z for $t \rightarrow \infty$.

Reverse SDE becomes

$$dY_t = \left(-f(T-t, Y_t) + \underbrace{g(T-t)^2 \nabla \log p_{X_{T-t}}(Y_t)}_{\text{score}} \right) dt + g(T-t) dW_t, \quad Y_0 \sim X_T.$$

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Approximation of score by a neural network s^θ :

$$\begin{aligned}\mathcal{L}(\theta) &:= \mathbb{E}_{t \sim \mathcal{L}_{[0,T]}} \mathbb{E}_{x \sim P_{X_t}} \left[\|s_t^\theta(x) - \nabla \log p_{X_t}(x)\|^2 \right] \\ &= \mathbb{E}_{t \sim \mathcal{L}_{[0,T]}, x_0 \sim X_0, x \sim P_{X_t|X_0=x_0}} \left[\|s_t^\theta(x) - \nabla \log p_{X_t|X_0=x_0}(x)\|^2 \right] + \text{const}\end{aligned}$$

Example: (continued)

$$P_{X_t|X_0=x_0} = \mathcal{N}(b_t x_0, (1 - b_t^2) I_d) \quad \text{with} \quad b_t = e^{-\frac{h(t)}{2}}$$

This implies

$$\nabla \log p_{X_t|X_0=x_0}(x) = \nabla \left(-\frac{1}{2(1-b_t^2)} \|x - b_t x_0\|^2 \right) = -\frac{x - b_t x_0}{1 - b_t^2}.$$

Plugging this into the loss function, we get

$$\mathcal{L}(\theta) = \mathbb{E}_{t \sim \mathcal{L}_{[0,T]}, x_0 \sim X_0, x \sim P_{X_t|X_0=x_0}} \left[\left\| s_t^\theta(x) + \frac{x - b_t x_0}{1 - b_t^2} \right\|^2 \right].$$

Once the score is computed, we can use it in the reverse SDE starting with the $Z \sim \mathcal{N}(0, I_d)$ which approximates X_T for T large enough.

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For the forward PDE

$$dX_t = f(t, X_t) dt + g(t) dW_t, \quad X_0 \sim P_{\text{data}},$$

the corresponding densities $p_t = p_{X_t}$ fulfill the **Fokker-Planck equation** (CE)

$$\begin{aligned} \partial_t p_t(x) &= -\nabla \cdot (f(t, x) p_t(x)) + \frac{1}{2} g(t)^2 \Delta p_t(x) \\ &= -\nabla \cdot \left(\underbrace{\left(f(t, x) - \frac{1}{2} g(t)^2 \nabla_x \log p_t(x) \right)}_{v_t} p_t(x) \right) \end{aligned}$$

Wasserstein gradient flow for minimizing $\mathcal{F} : \mathcal{P}_2(\mathbb{R}^d) \rightarrow \mathbb{R}$

Choosing $v_t \in -\partial \mathcal{F}(\mu_t) = -\nabla \delta \mathcal{F}(\mu_t)$ leads to the (CE)

$$\partial_t p_t = \nabla \cdot (p_t \nabla \delta \mathcal{F}(\mu_t))$$

◆ $\mathcal{F}_\nu(\mu) = \text{KL}(\mu, \nu) = \int_{\mathbb{R}^d} \log p(x) p(x) dx - \int_{\mathbb{R}^d} \log q(x) p(x) dx$, $\mu = p\lambda$, $\nu = q\lambda$
Then

$$\delta \mathcal{F}(p_t) = 1 + \log p_t - \log q, \quad \nabla \delta \mathcal{F}(p_t) = \frac{1}{p_t} \nabla p_t - \log q$$

so that

$$\partial_t p_t = \Delta p_t - \nabla \cdot (p_t \nabla \log q)$$

which is the above Fokker-Planck equation for $f(t, x) = \nabla \log q(x)$ and $g(t) = \sqrt{2}$.

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Fokker Planck Equation

For $q(x) = C^{-1}e^{-\beta\Psi(x)}$ we obtain the

Overdamped Langevin SDE:

$$X(0) = X^0 \sim p^0$$

$$dX_t = -\nabla\Psi(X_t)dt + \sqrt{2\beta^{-1}}dW_t$$

Euler-Maruyama forward step with step size η :

$$X_k := X_{k-1} - \eta\nabla\Psi(X_{k-1}) + \sqrt{2\beta^{-1}\eta}\xi_k, \quad \xi_k \sim \mathcal{N}(0, I_d)$$

Hint: Use substitution $s = \frac{t}{\beta}$, $\sqrt{\beta^{-1}}W_t = W_{t/\beta}$

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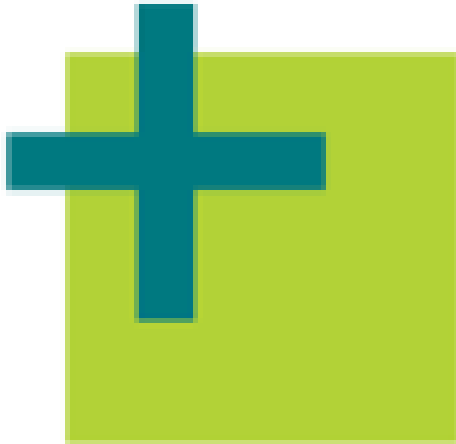
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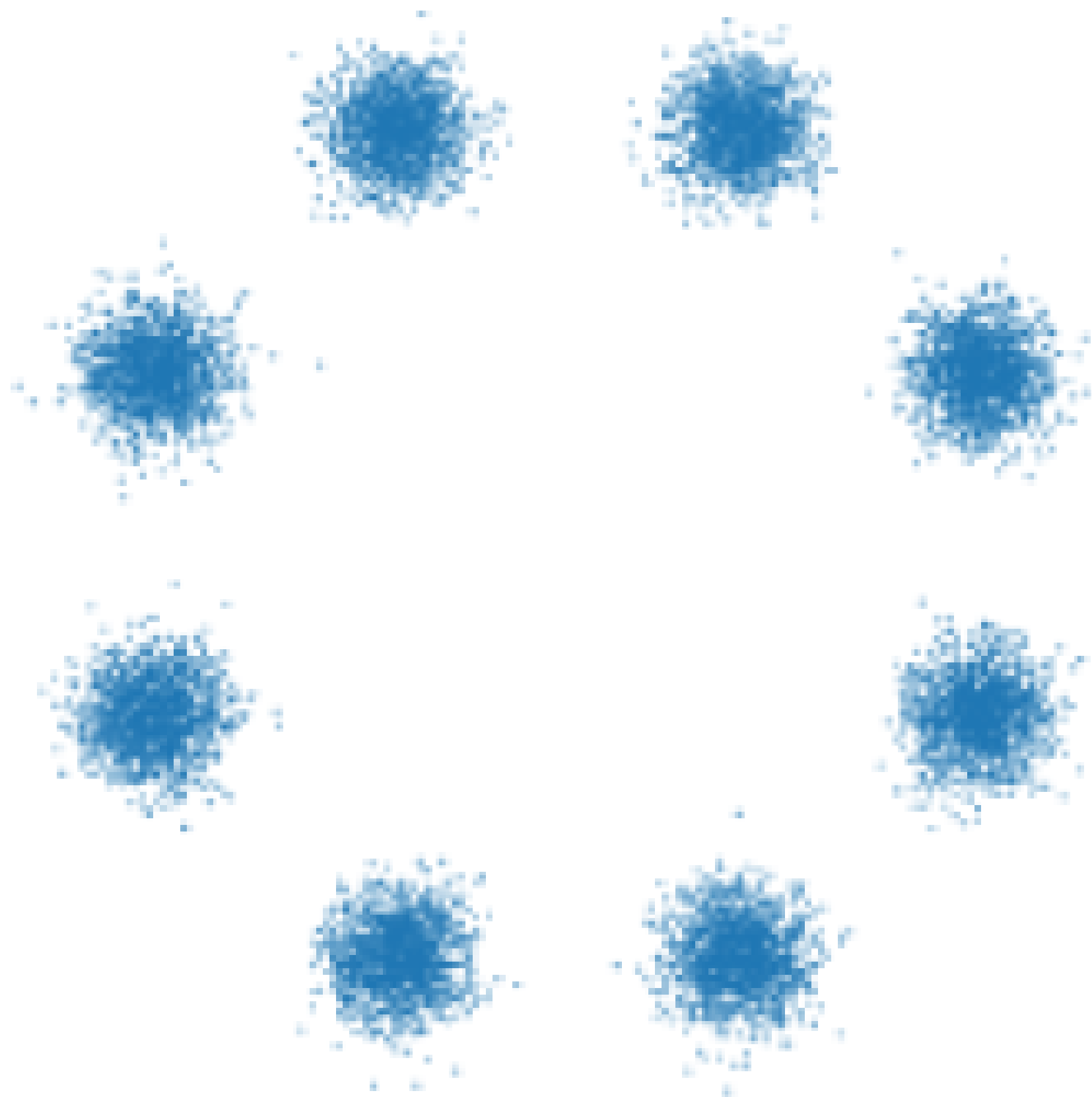
MATH 

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Variational and Information Flows in Machine Learning and Optimal Transport

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Bernhard Schmitzer
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François-Xavier Vialard
Christian Wald



3	4	2	1	9	5	6	2	1	8
8	9	1	2	5	0	0	6	6	4
6	7	0	1	6	3	6	3	7	0
3	7	7	9	4	6	6	1	8	2
2	9	3	4	3	9	8	7	2	5
1	5	9	8	3	6	5	7	2	3
9	3	1	9	1	5	8	0	8	4
5	6	2	6	8	5	8	8	9	9
3	7	7	0	9	4	8	5	4	3
7	9	6	4	7	0	6	9	2	3

